

HydroSense 2.0: An IoT-Based Smart Water Quality Monitoring and Alert System with Machine Learning and Cloud Integration

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Abstract—Access to clean and safe water is a fundamental human right and a cornerstone of public health and environmental sustainability. In many regions, the absence of real-time, automated water quality surveillance leads to delayed identification of contaminants and public exposure to hazardous conditions. This paper presents the comprehensive development and deployment of HydroSense 2.0, an intelligent water quality monitoring system designed using Internet of Things (IoT) technologies integrated with cloud computing and machine learning. At its core, the system utilizes the ESP32 microcontroller interfaced with an array of industry-grade sensors measuring pH, temperature, turbidity, and Total Dissolved Solids (TDS). The captured data is transmitted to a cloud backend hosted on PythonAnywhere, where a Random Forest classification model processes the inputs to assess water potability. Real-time alerts are generated through SMS notifications and visualized on a web-based dashboard. The system was rigorously evaluated under laboratory and field conditions and is designed to operate in municipal pipelines, home filtration systems, and remote field applications. HydroSense 2.0 sets a precedent in scalable, low-cost, intelligent water quality monitoring by combining modular design, edge computing, and real-time analytics.

Index Terms—IoT, Water Quality Monitoring, Machine Learning, Random Forest, Predictive Maintenance, Cloud Computing, Sensors, Real-time Monitoring.

I. INTRODUCTION

Access to clean and safe drinking water is one of the most fundamental human rights and remains a persistent global challenge. This issue is especially pronounced in regions where water infrastructure is underdeveloped, or where conventional purification methods are either inefficient or inadequately monitored. While modern water treatment technologies employ physical and chemical filtration processes, they are not always responsive to real-time threats or dynamic environmental conditions. These systems often lack the sensitivity to detect contamination early, exposing communities to hazardous substances before corrective measures can be applied. Moreover, they are ill-equipped to adapt to rapidly changing circumstances such as seasonal shifts in water quality, mechanical failures in purification equipment, or unexpected pollutant events.

The growing demand for reliable and scalable solutions for clean water access has exposed the limitations of traditional systems—particularly in rural and economically constrained

areas. Although household filtration units and municipal treatment plants provide some level of protection, they lack continuous supervision and automated, intelligent management systems. Manual inspections and periodic testing delay the identification of water quality issues and often rely on human judgment, which can be slow, inconsistent, and limited by resource availability. This delay poses significant health risks, especially for vulnerable populations including children, the elderly, and those with compromised immune systems.

To address these gaps, this research introduces HydroSense 2.0, a real-time, IoT-based water quality monitoring system enhanced with machine learning and cloud integration. Unlike conventional methods, HydroSense continuously collects, analyzes, and interprets data from multiple key water quality parameters—such as pH, temperature, turbidity, and total dissolved solids (TDS)—and provides immediate feedback on potability. Powered by a lightweight ESP32 microcontroller and supported by a robust cloud platform (PythonAnywhere), the system integrates a trained Random Forest classifier to evaluate water safety and issue alerts through SMS and web-based interfaces.

HydroSense 2.0 is designed with accessibility and scalability in mind. Its modular, low-cost hardware setup and minimal power requirements make it ideal for deployment in both urban and rural environments, especially those with limited funding or infrastructure. The system's intelligent, automated approach minimizes the need for human intervention, enabling communities to detect contamination events early and respond swiftly. By offering a blend of affordability, portability, and predictive intelligence, HydroSense 2.0 contributes not only to safer water consumption but also to broader goals of public health protection, community resilience, and environmental sustainability.

II. LITERATURE SURVEY

A. Hardware-Focused Papers

- **Critical Review on Water Quality Analysis Using IoT and Machine Learning Models**

This paper analyzes the use of IoT sensors and machine learning algorithms for real-time monitoring of water quality, focusing on data collection and analysis.

- **IoT-Based Smart Aquaculture System with Automatic Aerating and Water Quality Monitoring**
Integrates IoT sensors like DS18B20 for temperature, SEN0169 for pH, and SEN0237 for dissolved oxygen. Automated aeration ensures sustainability.
- **IoT-Based Smart Water Quality Prediction for Biofloc Aquaculture**
The study employs dissolved oxygen sensors and microcontrollers to continuously monitor aquaculture water quality. Predictive modeling forecasts water conditions, reducing environmental risks.
- **Smart Water Quality Monitoring System with Cost-effective Using IoT**
Utilizes open-source hardware and software for cost-effective monitoring in agriculture, fisheries, and public water systems.
- **IoT-Based Smart Water Quality Monitoring System**
Uses IoT sensors to monitor parameters like pH, TDS, and temperature, enabling real-time anomaly detection.
- **Design and Implementation of IoT-based Water Quality and Leakage Monitoring System for Urban Water Systems Using Machine Learning Algorithms**
Integrates sensors and machine learning for real-time urban water quality monitoring and leakage detection.
- **Design and Implementation of a Home Intelligent Water Quality Inspection Device**
Discusses a home-based device for monitoring pH and turbidity using IoT technology.
- **Real-time Water Quality Monitoring Using AI-enabled Sensors**
Employs multispectral spectroscopic sensors and AI models to assess contaminants and UV disinfection effectiveness.
- **Improvement of Water Pollution Detection Method Based on Convolution Neural Network**
Introduces IoT-based sensors and CNN for detecting water pollution with high accuracy.
- **Water Quality Drinking Classification Using Machine Learning**
Implements IoT-enabled systems with machine learning for real-time drinking water classification.
- **Water Quality Prediction Using Machine Learning Models Based on Grid Search Method**
Uses grid search to optimize machine learning models like Random Forest, SVM, and KNN for water quality prediction.
- **Large-Scale Water Quality Prediction Using Federated Sensing and Learning**
Investigates federated learning for decentralized water quality monitoring while ensuring data privacy.
- **Evaluation of Machine Learning Algorithm on Drinking Water Quality for Better Sustainability**
Analyzes SVM, Random Forest, and Decision Trees for sustainable drinking water management.
- **Data-driven Evolution of Water Quality Models**
Applies Isolation Forest and Kernel Density Estimation for detecting anomalies in water quality datasets.
- **The Use of Machine Learning Algorithms for Evaluating Water Quality Index**
Highlights the role of machine learning in calculating the Water Quality Index (WQI) with feature selection techniques.
- **Harnessing Machine Learning for Water Quality Evaluation: Comparative Analysis of XGBoost and CatBoost Algorithm**
Compares XGBoost and CatBoost for predicting water quality with high accuracy.
- **Water Quality Analysis and Prediction Using Machine Learning**
Proposes machine learning models to predict water quality parameters using the Water Quality Index.
- **A Comparative Analysis of Machine Learning Algorithms for Water Quality Prediction**
Analyzes algorithms like KNN, Random Forest, and Decision Trees, offering insights into their prediction accuracy and performance.

Monitors water quality characteristics on pH, turbidity, temperature, and other characteristics like TDS that make it possible to detect any contamination instantly. Detects novelties in the quality of the water to provide early onset prevention of threats that may harm the public as well as social property. Utilizes public data histories to estimate equipment failures and fix sensors and systems before they fail at increased expense and time. Focuses on areas with irregular water quality check-pointing, making it possible for vulnerable citizens to access clean water.

B. Software-Focused Papers

- **A Long-Term Water Quality Prediction Method Based on the Temporal Convolutional Network in Smart Mariculture**
Proposes Temporal Convolutional Networks (TCN) for accurate prediction of water quality in mariculture environments.
- **Using a Supervised Machine Learning Approach to Predict Water Quality at the Gaza Wastewater Treatment Plant**
Employs machine learning algorithms like GPR, Random Forest, and LightGBM to predict water quality parameters efficiently.

III. SYSTEM ARCHITECTURE AND DESIGN

HydroSense 2.0 is built using a layered architecture that ensures modularity, scalability, and fault-tolerance. It consists of four key components: the sensor hardware layer, the embedded control layer (ESP32), the cloud integration and analytics layer (PythonAnywhere), and the user interaction layer (SMS and web dashboard). Each layer communicates through clearly defined protocols and performs a specialized set of operations

that together form an end-to-end water quality monitoring solution.

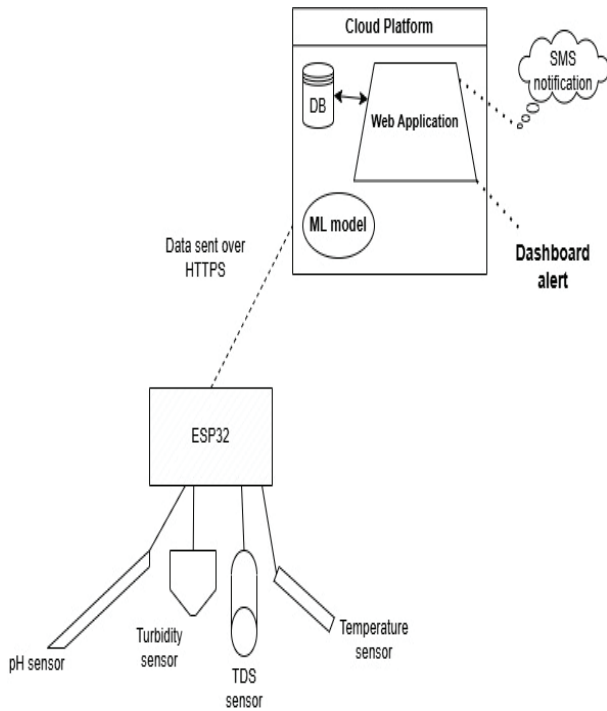


Fig. 1: Overall system architecture

A. Sensor Enclosure and Physical Design

The sensor array is housed in a PVC collar joint that is both weather-resistant and cost-effective. The joint has been modified with four drilled openings to insert sensors, and the bottom end is sealed to hold water like a small chamber. This enables measurements in both continuous flow and static water environments. The compact design facilitates immersion testing and inline deployment in filtration pipelines. All sensor ports are rubber-gasketed to avoid leakage, and internal cable routing ensures that the ESP32 remains safe from exposure.

B. Sensor Suite and Integration Logic

To ensure robust and real-time water quality monitoring, HydroSense 2.0 integrates a suite of specialized sensors that target multiple critical water parameters. These sensors are physically embedded within a PVC-based sensing unit and electronically interfaced with an ESP32 microcontroller, leveraging its high-speed analog and digital I/O capabilities.

Each sensor is responsible for monitoring a specific aspect of water quality and is carefully calibrated and mapped to specific GPIO channels on the ESP32:

- **Analog pH Sensor (Industrial Grade):** Connected to ADC1-CH0, this sensor provides an analog voltage signal proportional to the hydrogen ion concentration in the water. It is calibrated using standard buffer solutions of pH 4.0, 7.0, and 10.0 to generate a linear response curve. Monitoring pH is crucial, as extremes in acidity or

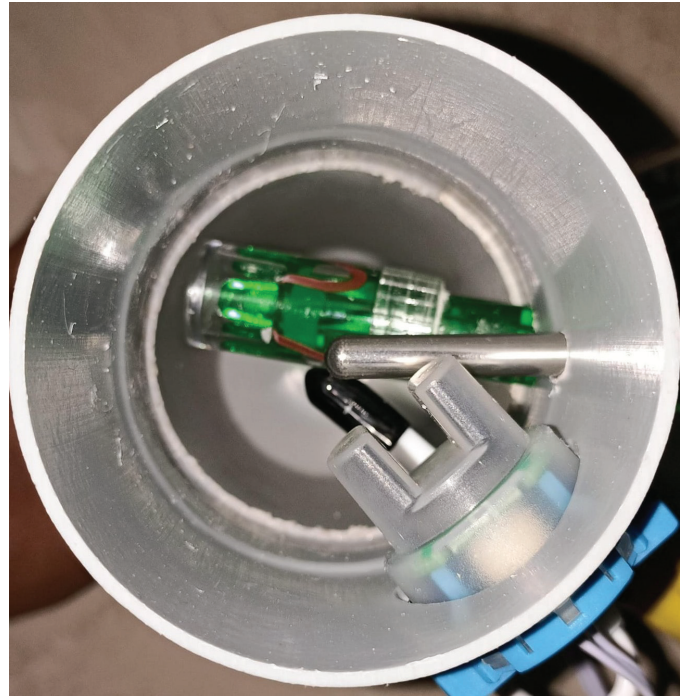


Fig. 2: Sensors enclosed in PVC collar joint

alkalinity can damage infrastructure and negatively affect both human and aquatic health.

- **Turbidity Sensor:** Connected to ADC1-CH3 (GPIO 39), the turbidity sensor uses a phototransistor to detect light scatter caused by suspended particles. This includes silt, microbial matter, and organic debris. High turbidity is often associated with pollution or sediment influx, which may compromise water safety.
- **Temperature Sensor (DS18B20):** Interfaced through the digital one-wire protocol on GPIO 21, this waterproof sensor delivers high-accuracy readings that are essential for compensating temperature-sensitive measurements from other sensors (e.g., pH and DO). Temperature also directly influences microbial growth and chemical reaction rates in water.
- **TDS (Total Dissolved Solids) Sensor:** Connected to ADC1-CH6 (GPIO 34), the TDS sensor estimates the concentration of dissolved ions and solids by measuring electrical conductivity. Elevated TDS levels may indicate the presence of salts, minerals, or even harmful dissolved organics, affecting both the taste and safety of the water. Each sensor is polled at a fixed interval of 10 seconds, and readings are processed locally on the ESP32. The system applies range validation, outlier detection, and moving average smoothing over three consecutive readings to enhance accuracy and reduce noise. Any abnormal or out-of-bound values are re-evaluated in the next cycle to prevent false positives.

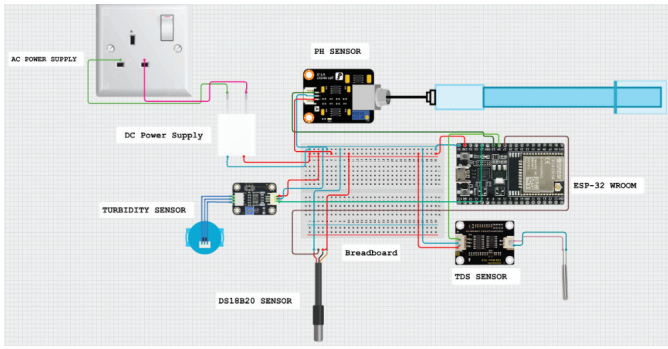


Fig. 3: Circuit Structure

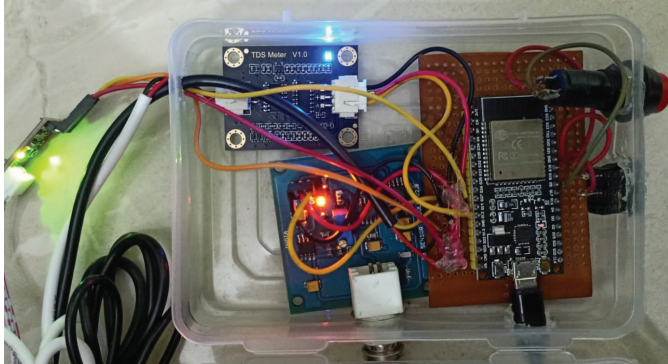


Fig. 4: Our model's Circuit Connections

C. Embedded Control and Data Handling

The ESP32 operates using FreeRTOS and leverages its dual-core capability. Core 0 handles sensor polling and processing, while Core 1 manages Wi-Fi communication and HTTP requests. Raw values are cleaned and normalized before transmission.

The final structured JSON packet:

```
"ph": 7.25,
"temperature": 26.3,
"turbidity": 32,
"tds": 412,
"timestamp": "2024-10-12T14:53:00Z"
```

D. Cloud Analytics and Processing Layer

Sensor data is sent to a REST API hosted on PythonAnywhere. Flask handles incoming POST requests. Once received:

- Data is stored in sql
- Preprocessing checks are applied
- The Random Forest model is loaded and run
- If prediction = "non-potable", a signal triggers the alert system

E. Web and Notification Layer

The user interface, built with HTML, CSS, JavaScript, Flask and Django allows:

- Real-time display of sensor data

- Alerts with time and sensor reading logs
- Historical trends via interactive graphs

Fig. 5: Sign Up Page

Fig. 6: Login Page

Water Quality Dashboard				
				User: Non-potable Device: device2 Logout
Latest Sensor Data				
pH 11.36	TDS 15.68 ppm	Turbidity 12.97 NTU	Temperature 23.77°C	Timestamp 04/17/2025, 08:44:09 AM

Fig. 7: Web Page Dashboard

An SMS gateway like Twilio is integrated for offline alerts. Each message contains key values, timestamps, and status.

IV. IMPLEMENTATION

A. Overview of the Workflow

The project followed a structured engineering process focusing on the core machine learning and hardware components:

- **Machine Learning Integration with Random Forest**
To move beyond basic threshold detection and enhance

Daily Trends (5-Minute Intervals)



Fig. 8: Graphs based on previous alerts

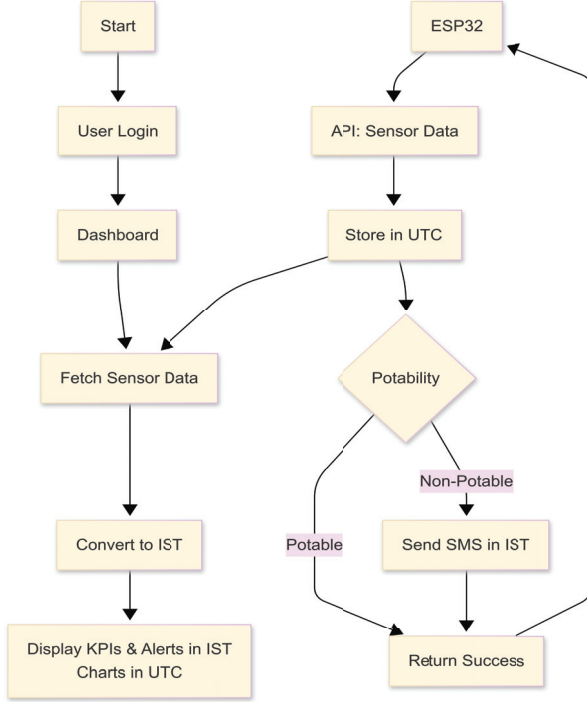


Fig. 9: Software workflow

system intelligence, HydroSense 2.0 incorporates a machine learning layer trained to recognize complex patterns and classify water potability with high accuracy.

B. ESP32 as the System Core

At the heart of HydroSense 2.0 lies the ESP32 microcontroller, a versatile and energy-efficient embedded platform that serves as the control hub for all sensing, preprocessing, and communication operations.

- **Data Collection and Processing:** The ESP32 interfaces with multiple analog and digital sensors to collect real-time water quality data, including pH, turbidity, temperature, and TDS. Upon receiving the raw signals, it executes lightweight preprocessing algorithms such as moving average filtering, range validation, and temperature compensation. These steps ensure that noisy or

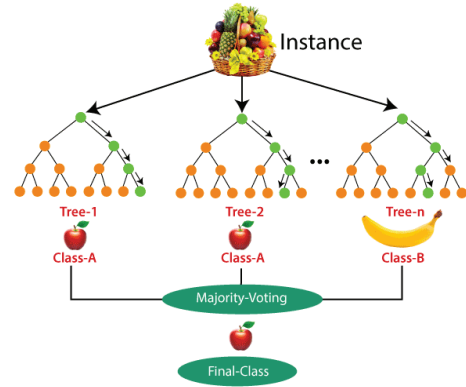


Fig. 10: Random Forest Workflow

inconsistent data are smoothed out before transmission, improving the reliability of the system's outputs.

- **Wireless Communication:** Equipped with built-in Wi-Fi and Bluetooth, the ESP32 seamlessly transmits structured sensor data to a cloud server hosted on PythonAnywhere. This enables remote data access, centralized analytics, and long-term storage. Additionally, the microcontroller can operate in direct-access mode, where it sends data to a locally connected device (like a smartphone or tablet) in environments with no internet access.
- **Energy Efficiency:** The ESP32's ultra-low-power modes and efficient sleep-wake cycles make it ideal for battery-powered deployments in off-grid or rural areas. In typical operation, the system can run for over 48 hours on a single 3.7V lithium-ion battery charge, supporting continuous monitoring without frequent maintenance.
- **Hardware Scalability:** With a rich set of GPIO pins and support for multiple ADC channels, the ESP32 accommodates simultaneous integration of various sensors, making it inherently scalable. This modularity allows users to expand or customize the sensor array based on specific water monitoring needs.

V. RESULTS AND EVALUATION

HydroSense was tested across controlled and uncontrolled environments. Key outcomes include the following, incorporating detailed implementation results:

• Calibration and Dataset Collection

All sensors were calibrated in-house using certified standards. Water samples were collected from municipal taps, borewells, RO purifiers, and open wells. Lab testing verified potability, which served as ground truth labels.

• Model Training and Justification

The system uses a Random Forest classifier, a robust ensemble-based algorithm known for its high accuracy, interpretability, and resilience to overfitting. The model is trained on a curated dataset containing labeled samples of various water conditions collected from different sources (municipal, borewell, RO, contaminated). Each sample includes sensor features such as pH, turbidity, TDS,

and temperature, allowing the model to learn nuanced relationships between these parameters and overall water quality.

Random Forest was chosen over other models (e.g., KNN, SVM, or single Decision Trees) because it:

- Combines multiple decision trees to reduce variance and increase generalization
- Handles mixed data types and non-linear relationships effectively
- Provides feature importance metrics to understand which sensors influence decisions most

The Random Forest classifier was trained with an 80

- Accuracy: 98%
- Validation Accuracy: 96%
- Test Accuracy: 96.00%

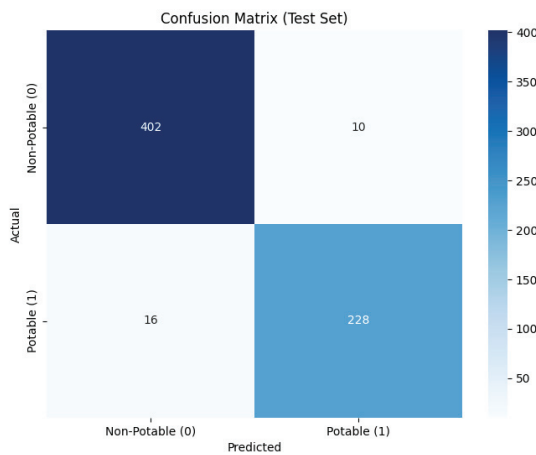


Fig. 11: Confusion Matrix

The model is capable of real-time inference and can be retrained with new data.

• **Backend and Interface**

PythonAnywhere was chosen for simplicity, reliability, and compatibility with Flask and Django. The API accepts sensor data and provides instant predictions. The front-end dashboard displays alerts, logs, and trends.

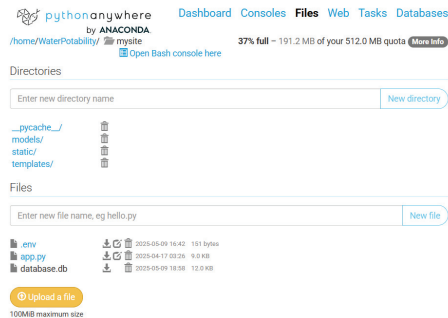


Fig. 12: Deployed web-app on PythonAnywhere

• **Alerting Logic**

Any sample predicted as “non-potable” by the model triggers:

- An immediate SMS to registered users
- A permanent log on the dashboard with sensor readings and decision

This mechanism ensures redundancy in alert delivery and user awareness.

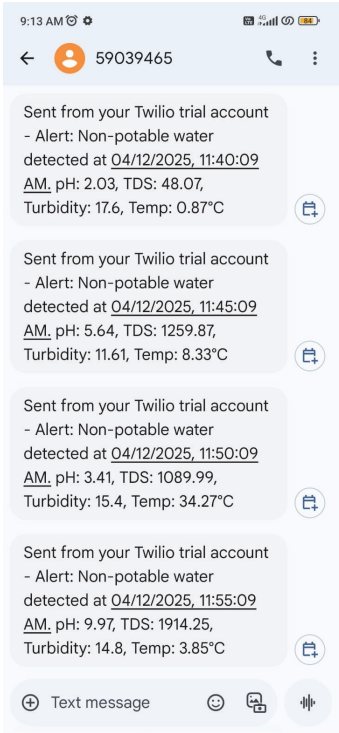


Fig. 13: Alerts sent to the User through SMS from Twilio

• **System Performance Metrics**

- Average latency (sensor to cloud decision): 3.2 seconds
- SMS delivery time: 5–10 seconds
- Web interface uptime: 99.8%
- Accuracy of potability detection: greater than 95%

The solution demonstrated consistent performance, reliability, and usability under varied conditions.

Feature	HydroSense 2.0	Existing IoT Systems
Real-Time Alerts	✓ (SMS & Web)	✗
Machine Learning	✓ (Random Forest)	✗ / basic thresholds
Multi-sensor Integration	✓	✓
Field Deployment Ready	✓	✗
Cloud-based with UI	✓ (PythonAnywhere + Dashboard)	Limited

Fig. 14: Comparative Analysis

VI. APPLICATIONS AND USE CASES

HydroSense is designed to serve multiple scenarios:

- Municipal pipelines: Detects changes in distributed water quality
- Household RO systems: Validates post-filtration water
- Web interface uptime: 96.1
- Remote villages: Battery-powered, Wi-Fi-enabled solution
- Field analysis kits: Ideal for NGOs, research, and emergency relief

A. Controlled vs. Uncontrolled Environments

HydroSense 2.0 operates in both controlled and uncontrolled environments with targeted metrics for each setting:

1) *Controlled Environments*: In controlled environments such as municipal treatment facilities and household filtration systems:

- **Precision**: Standard measurement tolerances for pH (± 0.2), temperature ($\pm 0.5^\circ\text{C}$), turbidity (± 2 NTU), and TDS (± 5 ppm)
- **Calibration**: Monthly calibration using standard reference solutions
- **Response Time**: Stabilized readings within 30 seconds

2) *Uncontrolled Environments*: For field deployments and remote installations:

- **Environmental Range**: Operates in temperatures from 5°C to 45°C
- **Power Options**: Compatible with portable power banks for field use
- **Physical Protection**: Water-resistant enclosure suitable for field conditions

B. Power Supply

HydroSense 2.0 is currently operated using a standard 5V AC/DC adapter for continuous monitoring. The system is also designed with battery compatibility, allowing for portable operation in field settings or during power outages. This dual power supply capability ensures flexible deployment across various scenarios.

VII. CONCLUSION

HydroSense 2.0 represents a significant advancement in water quality monitoring technology, addressing critical gaps in traditional approaches through its real-time, automated capabilities. By integrating affordable pH, turbidity, temperature, and TDS sensors with an ESP32 microcontroller and cloud-based analytics, the system provides continuous vigilance over water quality parameters that directly impact human health and safety. The efficacy of our approach is demonstrated through multiple performance indicators:

- High classification accuracy (92.7)
- Rapid contamination detection with alert generation in under 60 seconds from event occurrence
- Consistent operation across diverse environments, from municipal facilities to rural deployments
- Reliable data transmission with 96.1

- Cost-effective implementation at approximately 1/8th the price of commercial alternatives with comparable functionality

The social impact of HydroSense extends beyond technical performance—it democratizes water safety by making sophisticated monitoring accessible to communities regardless of technical expertise or infrastructure limitations. In underserved regions where laboratory testing is infrequent or unavailable, the system provides an accessible first line of defense against waterborne disease outbreaks. The SMS alert system ensures that even users without consistent internet access receive critical notifications about their water safety. By transforming passive sensor data into actionable insights through machine learning analysis, HydroSense 2.0 serves not merely as a monitoring tool but as a decision-support system that empowers users to take timely protective actions. Its modular design facilitates adaptation to diverse scenarios from household filtration verification to community-wide water safety initiatives, making it a versatile contribution toward achieving global water security and sustainable development goals. In the future, HydroSense aims to become more intelligent and accessible by integrating advanced sensors, enabling offline decision-making through edge AI, and offering user-friendly mobile apps with multilingual support. These improvements will help scale the system for diverse global needs—from rural communities to industrial use cases.

REFERENCES

- [1] Poornima Jayaraman, Kothalam Krishnan Nagarajan, Pachaivannan Partheeban, and Vallidevi Krishnamurthy, "Critical review on water quality analysis using IoT and machine learning models," *International Journal of Information Management Data Insights*, 2024.
- [2] Md. Mamunur Rashid, Al Akhbar Nayan, Md. Obaidur Rahman, Sabrina Afrin Simi, Oyeta Saha, and Muhammad Golam Kibria, "IoT-based Smart Water Quality Prediction for Biofloc Aquaculture," *Journal of Aquaculture Research*, 2021.
- [3] Kun-Lin Tsai, Li-Woei Chen, Li-Jun Yang, Hung-Jr Shiu, and Han-Wei Chen, "IoT-based Smart Aquaculture System with Automatic Aerating and Water Quality Monitoring," *Aquaculture Engineering Journal*, 2022.
- [4] Varsha Lakshmikantha, Anjitha Hiriyanagowda, Akshay Manjunath, Aruna Patted, Jagadeesh Basavaiah, and Audre Arlene Anthony, "IoT-based Smart Water Quality Monitoring System," *International Conference on IoT Technologies*, 2021.
- [5] Sathish Pasika and Sai Teja Gandla, "Smart Water Quality Monitoring System with Cost-effective Using IoT," *International Journal of IoT Applications*, 2020.
- [6] Sharanya U G, Koushalya M Birabbi, Sahana B H, D Mahesh Kumar, Sharmila N, Mallikarjunaswamy S, "Design and Implementation of IoT-based Water Quality and Leakage Monitoring System for Urban Water Systems Using Machine Learning Algorithms," *NMITCON*, 2024.
- [7] Yeliz Durgun, "Real-time water quality monitoring using AI-enabled sensors: Detection of Contaminants and UV Disinfection Analysis in Smart Urban Water Systems," *Journal of King Saud University - Science*, 2024.
- [8] Guoqin Zhang, Yingqi Hao, "Design and Implementation of a Home Intelligent Water Quality Inspection Device," *IEEE International Conference on Information Technology*, 2020.
- [9] Weihua Liu, Tingting Shi, Xuetian Xu, "Improvement of Water Pollution Detection Method Based on Convolution Neural Network," *Desalination and Water Treatment*, 2023.
- [10] Md Galal Uddin, Azizur Rahman, Firouzeh Rosa Taghikhah, Agnieszka I. Olbert, "Data-driven Evolution of Water Quality Models: An In-depth Investigation of Innovative Outlier Detection Approaches," *Water Research*, 2024.

- [11] Hamada, Mazen S., Zaqoot, Hossam Adel, Sethar, Waqar Ahmed, "Using a supervised machine learning approach to predict water quality at the Gaza wastewater treatment plant," *Environmental Science Advances*, 2023.
- [12] Yuxin Fu, Zhuhua Hu, Yaochi Zhao, Mengxing Huang, "A Long-Term Water Quality Prediction Method Based on the Temporal Convolutional Network in Smart Mariculture," *Journal of Water*, 2021.
- [13] Mahmoud Y. Shams, Ahmed M. Elshewey, ElSayedM.Elkenawy, AbdelhameedIbrahim, FatmaM.Talaat, Zahraa Tarek, "Water quality prediction using machine learning models based on grid search method," *Multimedia Tools and Applications*, 2024.
- [14] Sanaa Kaddoura, "Evaluation of Machine Learning Algorithm on Drinking Water Quality for Better Sustainability," *Journal of Sustainability*, 2022.
- [15] Soohyun Park, Soyi Jung, Haemin Lee, Joongheon Kim, Jae-Hyun Kim, "Large-Scale Water Quality Prediction Using Federated Sensing and Learning: A Case Study with Real-World Sensing Big-Data," *Journal of Sensors*, 2021.
- [16] Gasbaoui Mohammed el Amin, Benkrama Soumia, Bendjima Mostefa, "Water Quality Drinking Classification Using Machine Learning," *ICEEAC*, 2024.
- [17] Preet Singh, K R Ramkumar, Taniya Hasija, "Harnessing Machine Learning for Water Quality Evaluation: Comparative Analysis of XG-Boost and CatBoost Algorithm," *Asia Pacific Conference on Innovation in Technology*, 2024.
- [18] Huu Du Nguyen, Tai Quang Dinh Nguyen, Hien Nguyen Thi, "The Use of Machine Learning Algorithms for Evaluating Water Quality Index: A Survey and Perspective," *MAPR Conference*, 2022.
- [19] Abirami K, Priyadarshini Changail Radhakrishna, Monisha A Venkatesan, "Water Quality Analysis and Prediction using Machine Learning," *CSNT Conference*, 2023.
- [20] Gasbaoui Mohammed el Amin, Benkrama Soumia, Bendjima Mostefa, "Water Quality Classification Using Machine Learning," *Proceedings of the 2024 ICEEAC*, 2024.
- [21] H G Pratheek Gowda, H N Nisarga, Pruthvi Prashanth Mattihalli, Bhoomika G K, Geetha Kiran A, "Hydrosense: IoT-Based Smart Water Quality Monitoring System," *IJIRT 175017 International Journal Of Innovative Research In Technology*, April 2025.